LITERATURE REVIEW · READING PLAN · V2

Discriminative Neuron Analysis in BERT for AI-Generated Text Detection

Paweł & Miłosz · Updated March 2026 · 2–3 weeks remaining (~15 hrs/week)

MODEL	CONFIRMED FINDINGS	TARGET DIRECTIONS
Frozen DistilBERT (CLS token activations)	1,350 / 9,216 discriminative neurons · Non-overlapping UMAP regions · Generator-varying counts	Causal validation (primary) · Generalization · Linear representation (add-on)

WHAT CHANGED FROM V1

→ **Three-cluster finding dropped.** Could not be reproduced; likely a deadline artefact. All references to Cluster 0/1/2 as established results removed.

→ **U-shaped layer distribution removed** as a headline claim. Layer distribution varies per generator and is not characterised yet.

→ **Clustering papers (10, 11) dropped.** Dalvi et al. and Durrani EMNLP 2022 were primarily useful for interpreting the cluster structure.

→ **SAE / Monosemanticity papers (14, 16) dropped.** SAE direction deferred to a follow-up paper.

→ **Causal validation is now primary.** Papers 13, 17, and 20 upgraded to STUDY. New paper Enkhbayar 2025 added as critical cautionary reference.

→ **Detection landscape added.** RAID benchmark (Dugan et al.), M4 (Wang et al.), Kushnareva et al., Kuznetsov et al., Xia et al., and Zanotto & Aroyehun added as new papers.

→ **Project context updated.** Actual current findings: distinct non-overlapping regions in UMAP/PCA space; discriminative neuron counts vary across generators; layer distribution patterns differ per generator.

WEEK 1 — COMPLETED

Foundations

Papers 1–6 · ~15 hrs · Done

Conceptual vocabulary for all subsequent reading. No changes to this week — listed for record only.

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01 **A Practical Review of Mechanistic Interpretability** STUDY DONE 3.5 hrs

Rai et al., 2024 · arXiv / ICML 2025 tutorial · arxiv.org/pdf/2407.02646

Focus: §1–5 (Introduction through Beginner's Roadmap)

02 **A Primer in BERTology** STUDY DONE 3.0 hrs

Rogers, Kovaleva & Rumshisky · TACL 2020 · aclanthology.org

Focus: §2–5, §7

03 **Neuron-level Interpretation of Deep NLP Models: A Survey** STUDY DONE 2.5 hrs

Sajjad, Durrani & Dalvi · TACL 2022 · aclanthology.org

Focus: §1–5, §7

04 **Transformer Feed-Forward Layers Are Key-Value Memories** READ DONE 1.5 hrs

Geva et al. · EMNLP 2021 · aclanthology.org

Focus: §1–3

05 **BERT Rediscovers the Classical NLP Pipeline** READ DONE 1.0 hr

Tenney, Das & Pavlick · ACL 2019 · aclanthology.org

Focus: §1–4

06 **What Does BERT Look At? An Analysis of BERT's Attention** READ DONE 1.5 hrs

Clark et al. · BlackboxNLP @ EMNLP 2019 · arxiv.org/pdf/1906.04341

Focus: §2–4

WEEK 2 — CAUSAL METHODS + CORE DETECTION

Causal Methods + Core Detection

Papers 7–13 + Enkhbayar · ~13.5 hrs

Build the methodological vocabulary for causal validation and establish your closest paper relatives. The three STUDY papers here (08, 09, 10) are the load-bearing references for Direction A — read them as a methodologist, not just a reader.

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07 **Discovering Salient Neurons in Deep NLP Models** READ 2.0 hrs

Durrani, Sajjad & Dalvi · JMLR 2023 · jmlr.org/papers/v24/23-0074.html

Focus: §1–5

Closest methodological relative to your neuron identification pipeline. Uses correlation-based neuron selection similar to yours; §5 on layer distribution is the most directly comparable prior work.

08 **Confounding Neurons in Text Classification** STUDY 3.0 hrs

Borile & Abrate · Findings of EMNLP 2025 · aclanthology.org

Focus: All sections including appendices

Most directly related paper. Read it now primarily as a methodological template for activation patching — their intervention design (how they patch, what baselines they use, how they measure effect) is what you'll replicate. Conceptually, their confounding neurons are the opposite side of your discriminative neurons.

09 **Knowledge Neurons in Pretrained Transformers** STUDY ↑ UPGRADED 2.0 hrs

Dai et al. · ACL 2022 · aclanthology.org

Focus: §1–4

Now load-bearing for causal validation. Their ablation methodology — suppress identified neurons, measure accuracy drop, compare against random neuron baselines — is almost exactly what you'll implement. The 29% / 31% suppression / amplification finding is your primary empirical comparison point. Upgraded from READ because Direction A is now primary.

10 **Interpretability in the Wild: A Circuit for Indirect Object Identification in GPT-2 Small** STUDY ↑ UPGRADED 2.0 hrs

Wang et al. · ICLR 2023 · openreview.net

Focus: §1–3

The standard methodological reference for activation patching. Now primary for Direction A — the patching setup you'll implement follows this directly. Upgraded from READ because causal validation is now the core contribution rather than a validation add-on.

11 **Discriminative Neurons and the Paradox of Causal Necessity in Literary vs. AI Text** READ NEW 1.0 hr

Enkhbayar · arXiv 2025 · arxiv.org/abs/2510.17909

Focus: All sections (short paper)

Critical cautionary reference. Found that the top-50 discriminative neurons in GPT-2 were causally unnecessary — ablating them actually improved literary quality by 25.7%, concluding "correlation does not imply causal necessity." This was on a generative model, not a classifier; answering whether the same paradox holds in your setting is now an explicit research question your paper addresses. You must engage with this directly.

⚠ Note: arXiv preprint. Verify before final citation that it has not been withdrawn or substantially revised.

12 **Finding Neurons in a Haystack: Case Studies with Sparse Probing** READ 1.5 hrs

Gurnee et al. · TMLR 2023 · arxiv.org/pdf/2305.01610

Focus: §1–3, §5.1–5.2

Sparse probing as a methodological comparison — a cleaner version of your neuron identification approach. §5 case studies on monosemantic neurons and layer dynamics are useful for the neuron characterisation section regardless of whether you pursue corpus search.

13 **Pathway to Relevance: How Cross-Encoders Implement a Semantic Variant of BM25** STUDY ↑ UPGRADED 2.0 hrs

Lu, Chen & Eickhoff · EMNLP 2025, pp. 25525–25547 · aclanthology.org

Focus: §1–4

Activation patching applied specifically to BERT in a classification context — the closest methodological precedent for your causal analysis setup. Also a strong model for how to structure and write an interpretability-focused BERT paper. Upgraded from READ because you will follow their experimental design closely.

WEEK 3 — DETECTION LANDSCAPE + SYNTHESIS

Detection Landscape + Synthesis

Papers 14–25 · ~15 hrs

Cover the detection literature broadly enough to write related work and position your contribution precisely. Several papers here are new additions covering representation-based detection, interpretability of detectors, and benchmark papers for generalization. Most are READ or SKIM — this week is about breadth, not depth.

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14 **RepreGuard: Detecting AI-Generated Text via Hidden Representation Monitoring** READ ↓ DOWNGRADED 2.0 hrs

Chen et al. · TACL 2025, vol. 13 · doi.org/10.1162/TACL.a.61

Focus: All sections

Most similar paper in overall framing — uses hidden representations for AI text detection. Natural comparison point for your neuron-level analysis. Downgraded from STUDY because the cluster interpretation narrative it mainly supported is gone; it remains a key citation but not load-bearing methodology.

■ Verify section headings against the published TACL version.

15 **RAID: A Shared Benchmark for Robust Evaluation of Machine-Generated Text Detectors** READ NEW 1.5 hrs

Dugan et al. · ACL 2024, pp. 12463–12492 · aclanthology.org

Focus: §1–4, dataset construction section, evaluation results

This is your primary benchmark — an essential citation. Know exactly what's in the dataset: 6M+ generations across 11 models, 8 domains, 4 decoding strategies, 11 adversarial attacks. Reading this tells you precisely which generators you're analysing and grounds your generalization claims.

16 **Artificial Text Detection via Examining the Topology of Attention Maps** READ NEW 1.5 hrs

Kushnareva et al. · EMNLP 2021, pp. 635–649 · aclanthology.org

Focus: All sections

The closest published precedent for interpretability of a transformer-based AI text detector. Uses topological features of BERT attention maps plus probing analysis to mechanistically explain what the detector picks up. Directly relevant to the gap your paper fills — and the probing experiments are useful for framing your neuron interpretation claims.

17 **Robust AI-Generated Text Detection by Restricted Embeddings** READ NEW 1.5 hrs

Kuznetsov et al. · Findings of EMNLP 2024, pp. 17036–17055 · aclanthology.org

Focus: §1–4

Investigates which subspaces of BERT/RoBERTa embeddings carry genuine detection signals versus domain confounders, using head-wise PCA decomposition and concept erasure. Dual relevance: representation-based detection and interpretability. Provides a geometric representation-space analog to your neuron-level analysis — both ask which internal components matter for detection.

18 **Explaining Generalization of AI-Generated Text Detectors Through Linguistic Analysis** READ NEW 1.0 hr

Xia, Stańczak & Roth · EACL 2026, pp. 6524–6546 · aclanthology.org · arXiv: 2601.07974

Focus: §1–4, results

Bridges interpretability and generalization — explains detector generalization through 80 linguistic feature correlations across training and test conditions. Your paper does the equivalent at the neuron level, making this a natural contrast and companion. Useful for grounding any linguistic characterisation of what your discriminative neurons respond to.

19 **Text Fluoroscopy: Detecting LLM-Generated Text through Intrinsic Features** READ 1.5 hrs

Yu et al. · EMNLP 2024 · aclanthology.org

Focus: All sections

Uses layer-wise representation changes for AI text detection. Their AUROC-peaks-at-layer finding is the nearest analogue to your layer-specific discriminativeness results. Closest representation-based detection paper beyond RepreGuard.

20 **Exploring the Interpretability of AI-Generated Response Detection with Probing** READ 0.75 hrs

Choi & Zu · AIME-Con 2025 (ACL-affiliated workshop) · aclanthology.org

Focus: All sections (~6–8 pages)

Most directly related work at the specific intersection of detection and interpretability. Small workshop paper — read quickly. Main value is knowing what's been attempted at this intersection so you can claim the gap precisely.

■ Section headings inferred from abstract — verify against actual paper.

21 **Linguistic and Embedding-Based Profiling of Texts Generated by Humans and Large Language Models** SKIM NEW 0.75 hrs

Zanotto & Aroyehun · EMNLP 2025, pp. 22852–22869 · aclanthology.org

Focus: §1–3 only, then results tables

Most systematic characterisation of linguistic differences between AI and human text at a top venue — 8 domains, 11 LLMs, multiple linguistic levels. Useful for grounding interpretations of what your discriminative neurons might encode and for providing linguistic context in discussion.

22 **M4: Multi-generator, Multi-domain, and Multi-lingual Black-Box Machine-Generated Text Detection** SKIM NEW 0.75 hrs

Wang et al. · EACL 2024, pp. 1369–1407 · 🏆 Best Resource Paper · aclanthology.org

Focus: §1–3, generator/domain overview table

Second major multi-generator benchmark alongside RAID. Demonstrates systematic cross-generator × cross-domain generalisation failures in detectors — motivates why your neuron-

level generalization analysis matters. Know it exists and how it differs from RAID; one representative citation.

23 **A Survey on LLM-Generated Text Detection: Necessity, Methods, and Future Directions** SKIM 1.5 hrs

Wu et al. · Computational Linguistics 2025, pp. 275–338 · aclanthology.org

Focus: §1–2 in full → §3.3 (Neural-Based) only → §5–6 in full

Canonical detection survey. Read to know the landscape and correctly position your interpretability contribution within the broader detection field. Targeted skim — skip classification-heavy sections.

24 **DetectGPT: Zero-Shot Machine-Generated Text Detection using Probability Curvature** SKIM 0.75 hrs

Mitchell et al. · ICML 2023 · mlr.press

Focus: §1–2 only

Canonical zero-shot detection reference — one paragraph in related work contrasting the statistical paradigm with your representation/neuron approach. Read just enough to cite it correctly.

25 **Open Problems in Mechanistic Interpretability** SKIM 1.0 hr

Sharkey et al. · TMLR September 2025 · arxiv.org/abs/2501.16496

Focus: §1–2 only; read §2.1.2–2.1.4 carefully

Frame your contribution in terms of what the MI community considers unsolved. Useful for introduction and discussion. Citation note: fully published TMLR September 2025 — not a preprint.

Suggested Schedule

Weeks 2–3 · Up to 15 hrs/week

Grouped by natural reading order. Adjust days freely — the groupings matter more than exact day assignment.

BLOCK	PAPERS	EST. HOURS
WEEK 2 — Causal Methods + Core Detection		
Block 1	07 (Salient Neurons) + start 08 (Confounding Neurons)	3.5
Block 2	08 continued (finish) + 09 (Knowledge Neurons)	3.5

BLOCK	PAPERS	EST. HOURS
Block 3	10 (IOI Circuit) + 11 (Enkhbayar causal paradox)	3.0
Block 4	12 (Sparse Probing) + 13 (Pathway to Relevance)	3.5
Buffer	*Catch-up / notes comparison with Miłosz*	—
WEEK 3 — Detection Landscape + Synthesis		
Block 5	14 (RepreGuard) + 15 (RAID benchmark)	3.5
Block 6	16 (Attention Topology) + 17 (Restricted Embeddings)	3.0
Block 7	18 (Explaining Generalization) + 19 (Text Fluoroscopy) + 20 (Probing workshop)	3.25
Block 8	21 (Linguistic Profiling) + 22 (M4) + 23 (Detection Survey) + 24 (DetectGPT) + 25 (Open Problems)	4.75
After Week 3	Draft related work skeleton	—
TOTAL (Weeks 2–3)		**~28.5 hrs**

Post-Lit-Review Directions

Updated for current paper state

Three-cluster finding dropped. Directions revised accordingly. Commit to A + B at minimum; C is cheap enough to add without significant overhead.

DIRECTION A — PRIMARY

Causal Validation: Ablation + Activation Patching

Verify that discriminative neurons *causally drive* classification, not merely correlate with it. Ablation (zero out discriminative neurons, measure accuracy drop vs. random baseline) answers necessity. Activation patching (substitute AI-text activations into human-text inference at discriminative neuron positions) answers sufficiency and direction of effect. Together these move the paper from descriptive to mechanistic — the difference between a workshop paper and a main venue contribution.

Critical question to answer: does the Enkhbayar paradox (discriminative neurons causally unnecessary in a generative model) replicate in a classifier? Either result is publishable and framed correctly either way.

Compute: ~5–10 A100 hrs

Methods refs: Papers 09, 10, 13 (Dai, Wang, Lu)

Effect: Elevates core contribution substantially

DIRECTION B — RECOMMENDED

Generalization Across RAID Generators

Analyse which discriminative neurons are shared across generators and which are generator-specific. Overlap analysis across all 11 RAID generators plus cross-generator transfer test (train on neurons from generator A, classify generator B text). Either outcome is novel and publishable: shared neurons suggest a universal AI-text signal; non-overlapping neurons suggest generator-specific fingerprints detectable at neuron level.

Compute: Minimal (frozen model, sklearn)

Methods refs: Paper 15 (RAID), Paper 22 (M4)

Effect: Preempts key reviewer objection

DIRECTION C — CHEAP ADD-ON

Linear Representation Analysis

Fit a linear direction (logistic regression weight vector or mean-difference vector) in CLS activation space separating AI from human text. Check whether discriminative neurons are the ones whose axes most closely align with this direction. If yes: geometric claim — discriminative neurons *define* the AI-ness direction, not just correlate with classification. Fast to run, gives interpretive depth that neither causal validation nor generalization alone provides.

Compute: Negligible (one matrix op)

Methods refs: Papers 17, 18 (Kuznetsov, Xia)

Effect: Answers "what do they encode geometrically?"

DIRECTION D — DEFERRED

SAE Analysis

Train a sparse autoencoder on discriminative neuron activations to decompose them into monosemantic features. More ambitious — requires training and significant compute. Better scoped as a second paper once the current one is submitted.

Compute: Significant

Effect: Follow-up paper territory

Dropped Papers

Removed from v2 reading list

DROPPED — REASONS NOTED

- **Dalvi et al., "Discovering Latent Concepts Learned in BERT" (ICLR 2022)** — was in v1 as Paper 10. Primary relevance was interpreting the three-cluster structure; that finding is dropped.
- **Durrani et al., "On the Transformation of Latent Space in Fine-Tuned NLP Models" (EMNLP 2022)** — was in v1 as Paper 11. Core-linguistic vs. task-specific neuron framing mapped directly to Cluster 1 vs. Cluster 2; no longer load-bearing.
- **Cunningham et al., "Sparse Autoencoders Find Highly Interpretable Features in Language Models" (ICLR 2024)** — was in v1 as Paper 14. SAE direction deferred to follow-up paper.

- **Bricken et al., "Towards Monosemanticity" (Transformer Circuits Thread, Oct 2023)** — was in v1 as Paper 16. Superposition and feature splitting were useful framing for cluster interpretation; less load-bearing without that narrative.
- **Li et al., "Echoes of BERT"** — removed in v1 (ICLR 2026 submission, subsequently withdrawn). Do not cite.

Citation Corrections

Carried forward from v1

INCORRECT CITATION	CORRECT CITATION
"Finding Neurons in a Haystack" — ICLR	TMLR 2023 (not ICLR)
Lu et al. — short title only	"Pathway to Relevance: How Cross-Encoders Implement a Semantic Variant of BM25"
"Open Problems in MI" — preprint	TMLR September 2025 (fully published)
"Towards Monosemanticity" — conference	Transformer Circuits Thread, October 2023 (web article, not conference)
Rai et al. — conference paper	arXiv preprint; associated ICML 2025 tutorial (not a conference paper)

Reading Levels — What They Mean

STUDY

Full engagement. Read every section in scope, take structured notes, write a one-paragraph summary from memory afterward. You should be able to explain the core methodology, main finding, and how it relates to your paper without looking at the text. Used for papers that directly

READ

Active reading of the sections listed in the Focus field. Understand the argument and methodology well enough to summarise it and cite it accurately. Take brief notes on the one or two things most relevant to your paper. You don't need to master every detail, but you should not

SKIM

Abstract, introduction, conclusion, and any figures or tables that look directly relevant — usually 15–30 minutes. Goal is landscape awareness: know what the paper claims, where it sits in the field, and how to cite it correctly in one sentence. You are not expected to

define your methods or establish your closest prior work.

be surprised if a reviewer asks about it.

have read the methods or experiments in detail.

Working notes protocol (carry forward from v1): For each STUDY and READ paper, Paweł and Miłosz write a one-paragraph summary independently, then compare. Divergences flag conceptual misalignments early, before they become problems in the paper draft. For SKIM papers, a single sentence summary is sufficient.